

Houston

On-Device Mars Habitat AI Controller

GDG AI Hack 2026 - MSI Cut the Cord - Team PoliSa - Repository: github.com/landigf/gdgaihack-2026

single MacBook M3 Pro, 18 GB unified

100% local inference

30 NASA PDFs / 1,292 FAISS chunks

0 outbound packets

airplane-mode verified

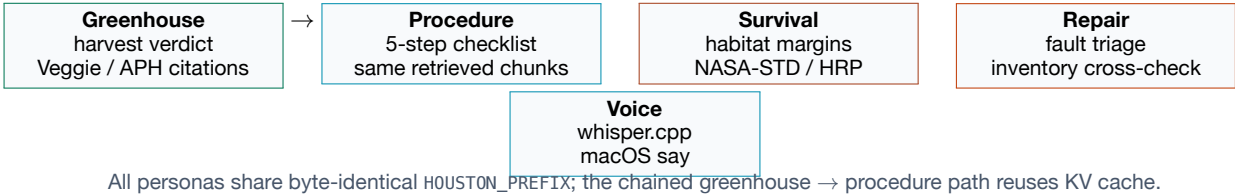
1. Architecture and product surface

Houston is a local mission-control assistant for a Mars surface habitat. Mars is the stress test where local AI is the correct architecture: Earth-Mars one-way light time is 4–24 minutes and solar conjunction can produce multi-day blackouts, so a cloud SLA cannot reach the operator in time.



Shell	Tauri 2 desktop app with native folder picker, PDF opening, microphone permission, React 18 dashboard and Three.js drill-ins.
Sidecar	Python FastAPI mounted on localhost only. Houston adds /ares/houston/*, /ares/sensor/query, /ares/perf without breaking Rover Core search/index/summarize.
Local AI	MLX-LM Qwen2.5-3B-4bit for generation; Ollama gemma3:4b fallback; Ollama nomic-embed-text; whisper.cpp ASR; macOS say TTS.
Knowledge	30 public NASA/space PDFs chunked into 1,292 FAISS vectors. Answers use structured [51] citations that open the source PDF.

2. Multi-agent design decisions



Why Mars	It turns “offline” from a privacy claim into a physics requirement: the cloud cannot answer during the operator’s decision window.
Why small models	On an 18 GB M3 Pro, a fast 3B 4-bit model plus RAG, schemas and streaming beat a larger model that wastes the latency budget.
Why citations	A habitat controller cannot hallucinate. Houston retrieves NASA chunks first and either cites them or returns a deterministic fallback.
Why app integration	The useful product is not a chatbot: it ties local files, PDF receipts, mic/TTS, sensor cache, 3D drill-ins and app state together.

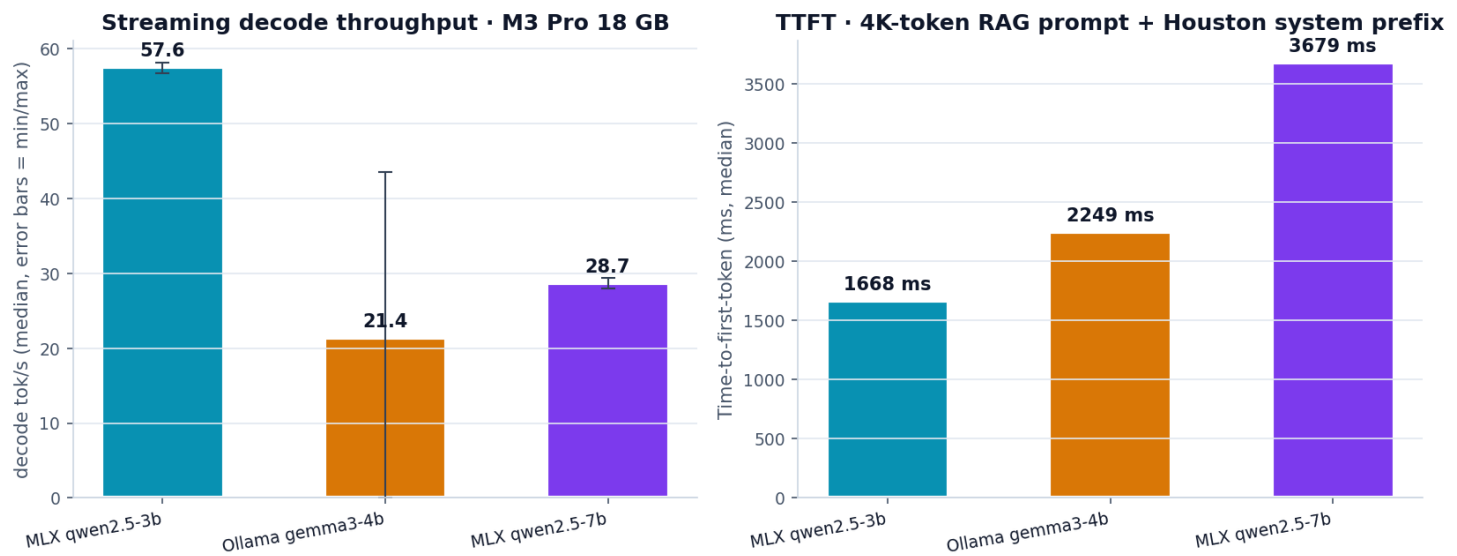
3. What we tried and learned

- MLX in-process is the operating point for this machine.** On the committed RAG-prefill benchmark, MLX Qwen2.5-3B hit **57.6** tok/s and **1,668** ms TTFT. The same plot keeps the slower local alternatives visible instead of hiding them.
- Cache the workflow, not only the model.** Shared persona prefix gave a measured **2.0x** A2A speedup; the Arrow/Parquet tile cache gave **29.2x** speedup on time-window sensor queries.
- Receipts matter on stage.** The PDF is not asking judges to trust us: every number below is regenerated from committed CSVs under benchmarks/houston/out/light/.

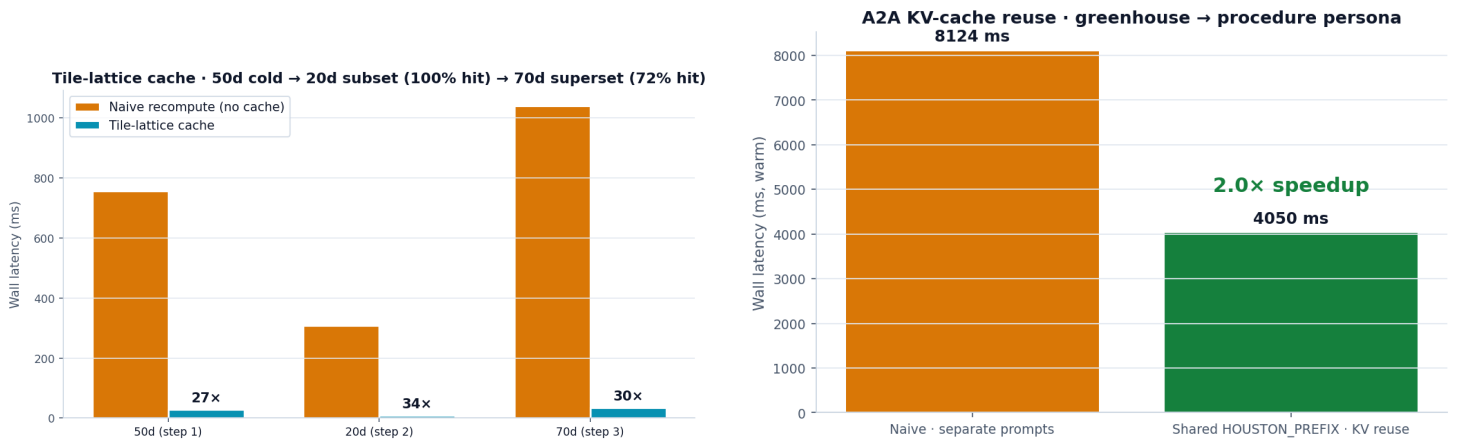
4. Compliance and reproducibility

At demo time Houston makes no hosted AI calls: no OpenAI, Anthropic, Google, Groq, Together, Replicate, cloud embeddings, cloud ASR or cloud TTS. Runtime services are local-only: Tauri app, FastAPI sidecar, MLX/Ollama, FAISS, whisper.cpp and macOS say. We verified airplane mode with scripts/airplane-check.sh and a 90-second tcpdump capture showing zero outbound packets outside localhost. Reproduce with ./scripts/setup.sh → ./scripts/download-mars-corpus.sh → LLM_BACKEND=mlx npm run tauri:dev.

5. Benchmark receipts from real plots

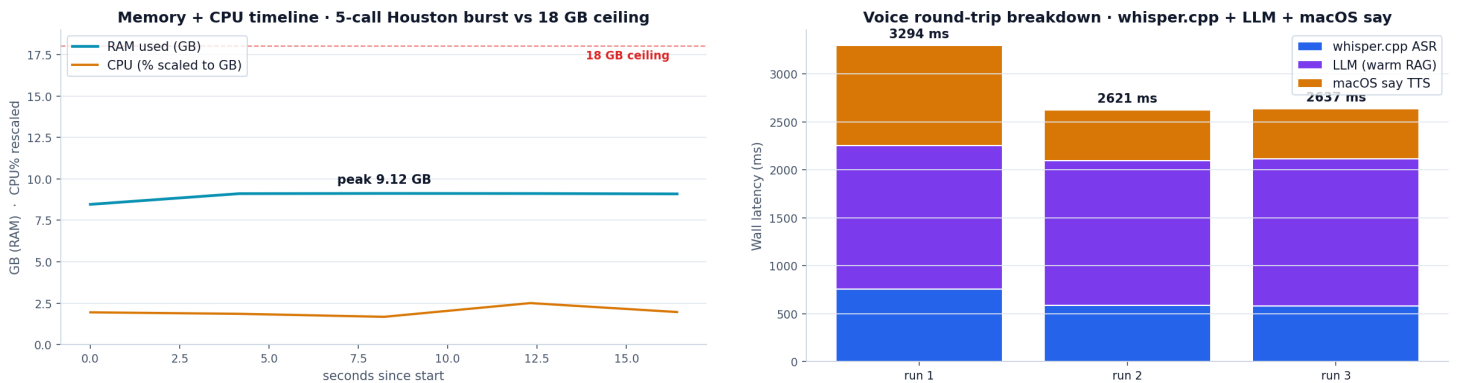


Throughput + TTFT. Three local generation backends on the same RAG-prefill prompt. Operating point: 57.6 tok/s and 1,668 ms TTFT on MLX Qwen2.5-3B-4bit.



Tile-lattice cache. 50d cold → 20d subset → 70d superset: 2,104 ms → 72 ms = 29.2x.

A2A KV-cache reuse. Greenhouse persona chains into procedure persona: 8,124 ms → 4,050 ms = 2.0x.



Memory and CPU. Five-call greenhouse burst stays below the 18 GB ceiling: about 9.1 / 18 GB peak RAM, zero swap.

Offline voice loop. whisper.cpp + LLM + macOS say: about 2.6s warm round-trip.

What we would add next: retrieval precision@k audit, reranker ablation, fresh-PDF indexing throughput, and an injection-safety test suite. We did not claim those numbers because they were not measured during the hackathon. The quality gate we did ship is citation grounding: Houston must answer from local [S_n] chunks or fall back safely.